



Social/Apartment Management System

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Abstract

- A web-based system for managing apartment residents, maintenance, complaints, and community activities.
- It improves communication between residents, admins, and technicians through a centralized platform.

Problem Statement & Objectives

- Problem:** Manual apartment management causes delays in complaints, maintenance, and resident communication.
- Objective 1:** Manage resident and apartment information digitally.
- Objective 2:** Track maintenance requests and complaints efficiently.
- Objective 3:** Improve communication between residents, admins, and technicians.
- Scope:** Residents, admins, technicians, apartments, complaints, maintenance, and notifications.

Existing vs Proposed System

Feature	Existing System	Proposed System
Resident Records	Paper/manual	Digital records
Complaints	Phone/paper	Online complaint system
Maintenance	Manual tracking	Request tracking
Communication	Phone/WhatsApp	Centralized platform
Limitation / Gap	Delays and poor tracking	Faster and organized management

Feasibility Study

- Technical:** React, Node.js/Express and MySQL provide a suitable web stack.
- Economic:** Open-source stack and free development tools reduce project cost.
- Operational:** Simple web interface for residents, admins, and technicians; reduces phone calls and paper records.
- Schedule:** Modules can be developed and tested incrementally.

Results & Testing

- Resident and apartment details are stored successfully.
- Complaints can be created and tracked.
- Maintenance requests can be assigned to technicians.
- Admin can manage residents and requests.

Conclusion

- Successfully developed a centralized apartment management system.
- The system reduces manual work and improves communication and service tracking.

Future Scope

- Online maintenance and utility bill payments.
- Push notifications for complaints and announcements.
- Visitor management and security integration.
- Mobile application for residents.

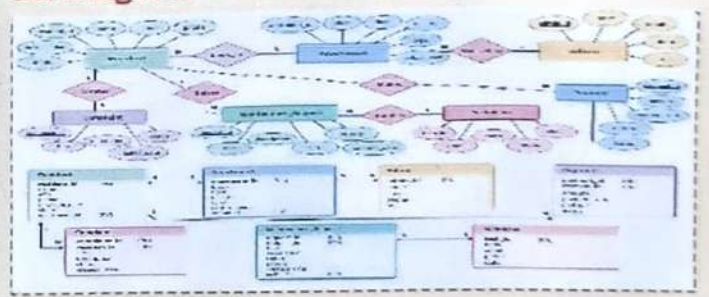
Tools & Technologies

- Front-End: HTML-CSS, JavaScript
- Back-End: Java, Spring-boot, Spring-Web
- Database: MongoDB
- Tools: Git & GitHub, Postman, VS Code

System Architecture



ER Diagram & Database Design



Implementation Screenshots

